

HAORAN GENG

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U.S. Permanent Resident

EDUCATION

- University of Notre Dame**, Notre Dame, IN 2021 – Present
Ph.D., Computer Science and Engineering (Expected 2026)
Advisor: Prof. Xiaobo Sharon Hu and Prof. Michael Niemier
- Columbia University**, New York, NY 2018 – 2021
M.S., Computer Engineering
- Rose-Hulman Institute of Technology**, Terre Haute, IN 2014 – 2018
B.S., Computer Engineering

RESEARCH EXPERIENCE

- Graduate Research Assistant**, University of Notre Dame Feb 2021 – Present
 - Conducting research in computer architecture and secure memory systems.
 - Leading the COSMOS project on RL-based counter cache optimization for secure memory.
 - Developing in-memory computing accelerators for security and encryption applications.
 - Published papers at MICRO, ICCAD, TCAD, and TVLSI.
- Research Assistant**, Columbia University Jan 2019 – May 2019
 - Worked under Prof. Henning Schulzrinne (IRT Lab).
 - Designed and built an IoT system using ML-based sound pattern classification.
- Research Collaborator**, Rolls-Royce, Indianapolis, IN Sep 2017 – May 2018
 - Developed automotive attack/defense systems for CAN BUS communication.
 - Implemented and validated techniques on a 2009 Honda Civic and a Tesla Model S.

TEACHING EXPERIENCE

- Teaching Assistant**, University of Notre Dame 2021 – Present
 - Assisted in undergraduate and graduate computer engineering courses.
 - Supported instruction, grading, and student project guidance.

PUBLICATIONS

Published and Accepted

- [1] **H. Geng**, X. Lu, Y. Che, Z. Tian, D. Cheng, X.-H. Sun, M. Niemier, X. Hu. *COSMOS: RL-Enhanced Locality-Aware Counter Cache Optimization for Secure Memory*. In *Proceedings of the 58th IEEE/ACM International Symposium on Microarchitecture (MICRO)*, 2025.
- [2] **H. Geng**, S. Wang, M. Li, T. S. Pereira, M. Niemier, X. Hu. *Towards Noise-Aware and Efficient Homomorphic Encryption on Crossbar Arrays*. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, accepted.
- [3] **H. Geng**, J. Mo, D. Reis, J. Takeshita, T. Jung, B. Reagen, M. Niemier, X. Hu. *PPIMCE: An In-Memory Computing Fabric for Privacy-Preserving Computing*. *Journal of Computer Science and Technology (JCST)*, 2025.
- [4] A. Mamdouh, **H. Geng**, M. Niemier, X. Hu, D. Reis. *Shared-PIM: Enabling Concurrent Computation and Data Flow for Faster Processing-in-DRAM*. *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems (TCAD)*, 2025. (Co-first author)
- [5] M. Li, **H. Geng**, M. Niemier, X. Hu. *Accelerating Polynomial Modular Multiplication with Crossbar-Based Compute-in-Memory*. In *Proceedings of the IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, 2023. (Co-first author)
- [6] D. Reis, **H. Geng**, M. Niemier, X. Hu. *IMCRYPTO: An In-Memory Computing Fabric for AES Encryption and Decryption*. *IEEE Transactions on Very Large Scale Integration Systems (TVLSI)*, 2021.

[7] T. S. Pereira, R. Wang, I. Nwozo, **H. Geng**, X. Hu. *Codebook-Based Acceleration for Ternary Language Model Inference*. Accepted to appear in the *Proceedings of the IEEE/ACM International Symposium on Low Power Electronics and Design (ISLPED)*, 2026.

SELECTED MANUSCRIPTS UNDER REVIEW

(First-author papers unless otherwise noted)

[8] **H. Geng**, Y. Che, D. Cheng, M. Niemier, X. Hu. *Secure Scattered Memory: High-Performance Memory Protection via Secret Sharing*. Submitted to *HPCA*, 2027.

[9] **H. Geng**, T. S. Pereira, X. Lu, X.-H. Sun, M. Niemier, X. Hu. *VIPER: A System-Level Performance Model for Exploring Processing-in-Memory Designs*. Submitted to *HPCA*, 2027.

PATENTS

[1] X. S. Hu, M. Niemier, **H. Geng**, et al. *Circuits and Systems for Programmable In-Memory Encryption/Decryption Fabric Including RA/CAM Arrays for Parallel Processing of Operations*. U.S. Patent No. 12,470,364, Nov. 11, 2025.

PRESENTATIONS

- *COSMOS: RL-Enhanced Locality-Aware Counter Cache Optimization for Secure Memory*, MICRO-58, Seoul, 2025.
- *Accelerating Polynomial Modular Multiplication with Crossbar-Based Compute-in-Memory*, ICCAD 2023, San Francisco, CA.
- Presented work on an in-memory computing accelerator for privacy-preserving computing, Techcon 2023, Austin, TX.

HONORS & AWARDS

- Third Place Award, DAC PhD Forum, 63rd ACM/IEEE Design Automation Conference (DAC), Long Beach, CA, 2026.

ACADEMIC SERVICE

- Reviewer: IEEE Transactions on Emerging Topics in Computing (TETC), 2024.

LEADERSHIP & OUTREACH

Undergraduate Research Mentor, University of Notre Dame *2025 – Present*

- Mentoring undergraduate research interns in in-memory computing and LLM.
- Leading students through experimentation and research development.
- Guiding student-led project toward submission to DAC.

PROFESSIONAL AFFILIATIONS

- IEEE student Member
- ACM student Member

PROFESSIONAL EXPERIENCE

Software Engineer Intern, OakGate Technology, Sacramento, CA *Jun 2019 – Sep 2019*

- Developed Python parsing scripts to enhance internal API functionality.
- Updated Bash scripts and C code for SSD device discovery debugging.
- Created NVMe CMB SSD conformance testing firmware in C.

REFERENCES

Prof. Xiaobo Sharon Hu, University of Notre Dame
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Prof. Michael Niemier, University of Notre Dame
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